

Applications of NMR Spectroscopy

1. Chemistry laboratories

Chemists rely on NMR Spectroscopy as a tool to chart the complex molecular structures of matter. The technique sees samples placed in a specialised NMR spectrometer, where it's exposed to a strong magnetic field. The gravitational pull generated by the field excites the nuclei of some atoms and creates resonant frequencies. These resonant frequencies are then tracked, measured and used to generate useful data.

2. Food quality control and research

Before hitting the consumer market, foods undergo strict quality control and research studies. NMR Spectroscopy is widely used across the industry to map protein structures, profile amino acids, identify carotenoids and quantify metabolites.

3. MRI scans

Most people are familiar with Magnetic Resonance Imaging (MRI) scans which use powerful magnetic fields and radio waves to reveal detailed images of the internal organs. Data is generated based on rates of absorption and energy emission recorded in the radiofrequency (RF) range. The medical imaging technique is founded in NMR Spectroscopy and uses the same basic principles.

4. Identifying human disorders

When used alongside metabolomics data, NMR Spectroscopy is an invaluable tool for identifying human disorders. The cellular metabolism contains important biomarkers used to diagnose a wide range of conditions, including tuberculosis, pneumonia and malaria. NMR Spectroscopy allows researchers to identify these tell-tale biomarkers and treat patients accordingly. The technique can also be used to investigate Parkinson's disease, as well as cardiovascular diseases and a range of neuropsychiatric disorders such as bipolar, schizophrenia, major depression and autism-spectrum disorders.

5. Cancer diagnosis

Over the past few decades, NMR Spectroscopy has become a key technique for cancer diagnosis. The ability to analyse abnormal behaviour in the cellular metabolism allows scientists to detect the metabolite-based biomarkers associated with cancers.

6. Environmental monitoring

It's not just the medical field where NMR Spectroscopy is considered a valuable tool, with the technique also used for environmental monitoring. Spectrometers are used to detect and characterise contaminants in air, soil and water samples as well as monitor the metabolic responses of organisms exposed to these contaminants.

7. Drug discovery and development

From trialling new cancer therapies to perfecting nutritional supplements, NMR Spectroscopy is a mainstay in the drug discovery and development arena.

Dr Robin J. Blagg, a representative from UK-based manufacturing and research company Oxford Instruments, offers more information on the latest NMR Spectroscopy techniques and instruments in 'X-Nuclei NMR Spectroscopy'.

NMR in Biomedical Research and development.

Nuclear magnetic resonance (NMR) is a spectroscopic technique to study molecules with NMR active nuclei. Not all nuclei can give NMR signals. Only those nuclei possessing non-zero nuclear spin have a nuclear magnetic moment which produces magnetic interactions with an external magnetic field. Sitting in an external magnetic field, the NMR active nucleus would have energy level splitting. A radio frequency pulse of the right frequency would induce transitions of nuclei from the lower energy levels to the higher energy levels. After the radio frequency pulse is switched off, the nucleus at the higher energy states would relax back to the lower states, giving out signals (figure 1). This is called the nuclear Zeeman Effect. Since the energy level splitting is different for different atomic nuclei, for a fixed external magnetic field, the frequency of the signal becomes a characteristic for a specific atom (Larmor frequency). Atoms such as ^1H , ^2H , ^{13}C , ^{15}N , ^{31}P , etc. are the common nuclei studied using NMR in biomedical research.

NMR is a powerful technique applied in a variety of field, especially in biomedical research. NMR can identify molecules, to probe molecular dynamics and study the molecular interactions. In biomedical research, it has been applied to the structure and function determination of important biological molecules, such as drug targets; study of drugs and the interactions with their targets; identification of abnormal metabolites as the biomarkers for specific diseases; diagnoses of disorders by imaging particular tissues or organs.

PROTEIN STRUCTURAL DETERMINATION USING SOLID-STATE NMR

Solid-state NMR is different from solution NMR in that the sample used in solid-state NMR is in the solid or semi-solid form. More than half of our body mass is water. Therefore, a lot of biochemical reactions take place in solution phase. However, there are still certain situations when the studies have to be carried out during the solid-state phase. There are solid tissues like bones, teeth, the structure of which could also be understood with the help of NMR. There are molecules embedded in the biological membranes. Membranes are in gel-like semi-solid form. Most protein drug targets are membrane proteins. They usually act as receptors or transporters to pass the information or molecules in and out of the membranes. Not only in human being, but there are also interesting solid-state protein structures in bacteria and viruses too. For example, the virus capsid (the virus shell) formed by capsid proteins has a unique structure with high symmetry. HIV capsid protein has been a hot topic in solid-state NMR field. The information gathered from the structural study will help us towards a cure for AIDS. Solid-state NMR is necessary to study those proteins while keeping them in the native environment.

PHARMACEUTICAL RESEARCH: DRUG SCREENING

With the development of technology, a huge number of low molecular weight compounds have been synthesized by the chemists and a large number of biological targets have been characterized by the biologist. These compounds and their derivatives are the potential drugs. High-throughput screening is needed to pick up the right pairs, the compound (as a ligand for its target) and its interacting target, often proteins or nucleic acids. Hopefully, this compound could also intervene with the action of the target in vivo, leading to drug discovery. The understanding of the structure of the target and the compound, the target-ligand complex and the interaction mechanism could help in improving the drug design. NMR is one of the tools that could be used here for the discovery.

CHEMICAL SYNTHESIS

NMR, especially solution NMR has been widely applied in chemical synthesis as an analytic tool to identify compounds, for example, chemical synthesis of QC-01–175, a tau protein degrader and biocatalytic manufacture of islatravir among others. Many organizations have built and maintained vast and comprehensive NMR spectra databases (e.g., www.acdlabs.com). NMR could also be a structure elucidation tool for those biological targets or target-ligand complexes, as discussed in the last section. Biological magnetic resonance databank is one of the places where a lot of structures of biological macromolecules are deposited. Furthermore in pharmaceutical research, NMR has also been developed to fast screen the potential drug compounds, lead generation and optimization, namely detecting the ligand binding to their biological macromolecular targets, locating the active sites, measuring the affinity or the kinetics of the binding and optimizing the structure of the ligands based on the results.

Applications of NMR Spectroscopy in Quality Control

Food Analysis by NMR:

NMR techniques are utilized in determining the changes on heat-treated foods. In this technique, change of water distribution and mobility in foods could be determined without damaging to food. Also, it is one of the techniques used to determine the level of melamine in foods. In addition, nuclear magnetic resonance spectroscopy is a good method to determine water holding capacity, intramuscular fat and total water content of meat.

- ❖ High resolution NMR technique can be used in the determination of organic compounds of vinegar and quantification of the main classes of vinegar components (carbohydrates, alcohols, organic acids, volatile compounds and amino acids).
- ❖ **Dairy products:** Successfully used another field of NMR spectroscopy is milk and dairy products. Studies were conducted on the water molecules of casein systems, serum proteins and milk powder by NMR technique. Furthermore, NMR applications are used to determine the characteristics of the cheese. Nuclear magnetic resonance is used to monitor the maturation and evaluate the modifications of amino acid and water content in cheeses.

Qualitative analysis Of Coffee by NMR:

Application of ^1H NMR was used quality control and authenticity of instant coffee by Charlton et al. The presence of inherent differences between coffees produced by different manufacturers, and even between those produced by the same manufacturer, by identifying 5-(hydroxymethyl)-2-furaldehyde as a marker compound using the structural characteristics were determined by NMR.

- ❖ Another study, ^{31}P NMR was used to determine the amount of mono- and diglycerides in virgin olive oils. It was found that quantification of other constituents of olive oils bearing functional groups with labile protons could be extended by quantitative ^{31}P NMR spectroscopy.
- ❖ **Foodomics:** NMR can be used for foodomics because of ease of quantification and identification, short time and low costs needed for analysis and high number of metabolites that can be measured through a single-pass. Because of highest sensitivity of NMR focus on hydrogen is preferred for foodomics studies.

Qualitative analysis Of Juices by NMR:

Discrimination between apple juices produced from different varieties has been achieved by applying principal components analysis (PCA) and linear discriminant analysis to ^1H NMR spectra of the juices by Belton et al. Under optimum conditions a 100% success rate was achieved. Examination of the principal component loadings showed that the levels of malic acid and sucrose were two important chemical variables, but variations in the composition of the minor constituents were also found to make a significant contribution to the discrimination.

Qualitative analysis Of Green tea by NMR:

Green teas from different countries was collected and analyzed by ^1H NMR. It was proposed to establish if the teas could be discriminated according to the country of origin or with respect to quality. After an extensive assignment of spectra, NMR spectroscopy has been shown to provide a wealth of information about the main metabolites of the teas studied. Tea components were determined for discrimination of teas.

Solid-state (NMR):

Solid-state magnetic ^{31}P nuclear resonance (NMR) to determine the different states of phosphates in cheeses was used. Sixteen semi-hard cheeses of various compositions were studied, and three fractions of phosphates (P) were distinguished according to their mobility: (1) mobile soluble P (ca. 10 % of total P), (2) mobile insoluble P (70 %) and (3) immobile insoluble P (20 %). In accordance with chemical composition and buffering capacities of the cheeses, these fractions could represent respectively (1) soluble inorganic P, (2) inorganic colloidal calcium P and phosphorylated serine residues (Pser) involved in a loose structure and (3) Pser involved in a tight environment. It was thus demonstrated that solid-state NMR is an appropriate method to observe the distribution of phosphates in cheese matrix and their evolution during cheese-making.

Measurement of conjugated linoleic acid:

The application of ^1H nuclear magnetic resonance (NMR) spectroscopy to the measurement of conjugated linoleic acid (CLA) content in the lipid fraction of dairy products is both a novel and inviting alternative to traditional methods such as gas chromatography (GC), which can require time-consuming sample derivatization. ^1H NMR analysis approach has potential application in the dairy industry as a screening technique for total CLA concentrations in large numbers of cheese samples and in the screening of CLA content in other dairy products.